

# TIANLIN LIU

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## EXPERIENCE

- Fall 2023      **Research Scientist Intern at Google DeepMind, Paris.**  
Aligning language models with human feedback [15, 16].  
Hosted by Mathieu Blondel, Felipe Llinares, and Daniele Calandriello.
- Summer 2022      **Research Scientist Intern at Google Brain, Zurich.**  
Improving vision mixture-of-experts models via optimal transport [9, 13].  
Hosted by Joan Puigcerver and Mathieu Blondel.

## EDUCATION

- 2020–PRESENT      **Ph.D. Computer Science**  
University of Basel, Switzerland  
Advisor: Ivan Dokmanić
- 2017–2019      **M.S. Data Engineering**  
Jacobs University Bremen, Germany  
Advisor: Herbert Jaeger
- 2013–2016      **B.S. Mathematics**  
Jacobs University Bremen, Germany  
Advisor: Götz Pfander

## AWARDS AND HONORS

- 2023      Google Conference Scholarship
- 2022      Young Researcher, Heidelberg Laureate Forum.
- 2022      Travel Award, SIAM Conference on Imaging Science.
- 2021      Oberwolfach Leibniz Graduate Student Grant, Oberwolfach Institute.
- 2019      Dean’s Award for Outstanding Master’s Thesis [4], Jacobs Univ. Bremen.
- 2019      IEEE NER 2019 Best Paper Finalist Award [2].
- 2018      SmartStart-1 Fellowship, Bernstein Network and Volkswagen Foundation.

## PUBLICATIONS AND MANUSCRIPTS

[Google Scholar](#)

(★ indicates equal contributions)

- [16] **Decoding-time Realignment of Language Models.**  
**Tianlin Liu**, Shangmin Guo, Leonardo Bianco, Daniele Calandriello, Quentin Berthet, Felipe Llinares, Jessica Hoffmann, Lucas Dixon, Michal Valko, Mathieu Blondel.  
arXiv Preprint, 2024.
- [15] **Direct Language Model Alignment from Online AI Feedback.**  
Shangmin Guo★, Biao Zhang★, **Tianlin Liu**, Tianqi Liu, Felipe Llinares, Bilal Piot, Johan Ferret, Alexandre Ramé, Thomas Mesnard, Yao Zhao, Mathieu Blondel.  
arXiv Preprint, 2024.
- [14] **Open RL Benchmark: Comprehensive Tracked Experiments for Reinforcement Learning.**  
Shengyi Huang★, Quentin Gallouédec★, ..., Xiao Hu, **Tianlin Liu**, Jongwook Choi, Brent Yi.  
arXiv Preprint, 2024.
- [13] **Routers in Vision Mixture of Experts: An Empirical Study.**  
**Tianlin Liu**, Mathieu Blondel, Carlos Riquelme, Joan Puigcerver.  
arXiv Preprint, 2024.
- [12] **GLIMPSE: Generalized Local Imaging with MLPs.**  
AmirEhsan Khorashadizadeh, Valentin Debarnot, **Tianlin Liu**, Ivan Dokmanić.  
arXiv Preprint, 2024.
- [11] **The N Implementation Details of RLHF with PPO.**  
Shengyi Huang, **Tianlin Liu**, Leandro Von Werra.  
The Blogpost Track of the International Conference on Learning Representations (ICLR), 2024.
- [10] **WaveBench: Benchmarking Data-driven Solvers for Linear Wave Propagation PDEs.**  
**Tianlin Liu**★, Jose Antonio Lara Benitez★, Florian Faucher★, AmirEhsan Khorashadizadeh, Maarten V. de Hoop, Ivan Dokmanić.  
Transactions on Machine Learning Research, 2024.
- [9] **Sparsity-Constrained Optimal Transport.**  
**Tianlin Liu**, Joan Puigcerver, and Mathieu Blondel.  
In Proc. of the International Conference on Learning Representations (ICLR 2023).  
**Spotlight presentation**
- [8] **Learning Multiscale Convolutional Dictionaries for Image Reconstruction.**  
**Tianlin Liu**, Anadi Chaman, David Belius, and Ivan Dokmanić.  
IEEE Transactions on Computational Imaging, 2022.
- [7] **Universal Approximation Under Constraints is Possible with Transformers.**  
Anastasis Kratsios, Behnoosh Zamanlooy, **Tianlin Liu**, and Ivan Dokmanić.  
In Proc. of the International Conference on Learning Representations (ICLR 2022).  
**Spotlight presentation**

- [6] **Finding Trainable Sparse Networks through Neural Tangent Transfer.**  
**Tianlin Liu** and Friedemann Zenke.  
 In Proc. of the International Conference on Machine Learning (ICML 2020).
- [5] **Causally Denoise Word Embeddings Using Half-Sibling Regression.**  
 Zekun Yang\* and **Tianlin Liu\***.  
 In Proc. of the AAAI Conference on Artificial Intelligence (AAAI 2020).
- [4] **Harnesing Slow Dynamics in Neuromorphic Computation.**  
**Tianlin Liu.** Master's thesis, 2019.  
**Dean's Prize for Outstanding Master's Thesis**
- [3] **Continual Learning for Sentence Representations Using Conceptors.**  
**Tianlin Liu**, Lyle Ungar, and João Sedoc.  
 In Proc. of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2019).
- [2] **Reservoir Transfer on Analog Neuromorphic Hardware.**  
 Xu He, **Tianlin Liu**, Fatemeh Hadaeghi, and Herbert Jaeger.  
 In Proc. of the International IEEE Conference on Neural Engineering (NER 2019).  
**Best Paper Finalist Award (3rd place out of 467 accepted papers)**
- [1] **Unsupervised Post-processing of Word Vectors via Conceptor Negation.**  
**Tianlin Liu**, Lyle Ungar, and João Sedoc.  
 In Proc. of the AAAI Conference on Artificial Intelligence (AAAI 2019).

## OPEN-SOURCE CONTRIBUTION

|                                                                |        |
|----------------------------------------------------------------|--------|
| <b>huggingface/transformers</b> : transformers library         | ☆ 120k |
| <b>huggingface/trl</b> : lightweight library for RLHF          | ☆ 7.3k |
| <b>PythonOT/POT</b> : python optimal transport                 | ☆ 2.2k |
| <b>google-research/vmoe</b> : vision mixture of experts models | ☆ 465  |
| <b>google/jaxopt</b> : differentiable optimizers in JAX        | ☆ 864  |

## JOURNAL AND CONFERENCE REVIEWING

|            |                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------|
| CONFERENCE | ICML (2021–2024), NeurIPS (2021–2023), ICLR (2022, 2023), LoG (2022)                                           |
| JOURNAL    | IEEE Transactions on Signal Processing, IEEE Signal Processing Letters, IEEE Open Journal of Signal Processing |

## TEACHING EXPERIENCE

|            |                                                       |                   |
|------------|-------------------------------------------------------|-------------------|
| 2022       | Mathematics of Data Science                           | Uni Basel         |
|            | Scientific Writing                                    | Uni Basel         |
|            | Machine Learning on Graphs, Groups, and Manifolds     | Uni Basel         |
| 2021       | Computational Sustainability and AI for Social Good   | Uni Basel         |
| 2021       | High-dimensional Data Analysis and Learning on Graphs | Uni Basel         |
| 2018, 2019 | Machine Learning                                      | Jacobs Uni Bremen |
| 2018, 2019 | Principles of Statistical Modeling                    | Jacobs Uni Bremen |